

MPS simulator produces incorrect result

<https://github.com/Qiskit/qiskit-aer/issues/1253>

ROW 99

MPS simulator produces incorrect result #1253

New issue



Closed



liyaochong opened on May 17, 2021

...

BUG TEMPLATE

Informations

- Qiskit AER version: 0.8.2
- Operating system: Windows

What is the current behavior?

The output of quantum circuit with 'matrix_product_state' and 'statevector' are different.
{'111000101001': 1024} for 'matrix_product_state' simulator;
{'100100011001': 1023, '100100110001': 1} for 'statevector' simulator.

Steps to reproduce the problem

```
import numpy as np
import qiskit
from qiskit import QuantumCircuit, transpile
from qiskit.providers.aer import AerSimulator
import networkx as nx
import matplotlib.pyplot as plt

n = 12
G = nx.Graph()
G.add_nodes_from(range(0,n))
G.add_edges_from([(0,1), (0,2), (1,2), (1,4), (2,4), (2,3), (3,5), (4,6), (5,6), (4,7), (6,7),
(7,8), (8,9), (7,9), (6,8), (6,9), (8,10), (9,11), (10,11)])
colors = ['b' for node in G.nodes()]
pos = nx.spring_layout(G)
default_axes = plt.axes(frameon=True)
nx.draw_networkx(G, node_color=colors, node_size=600, alpha=.8, ax=default_axes, pos=pos)

def produce_QAOA_state(gammas,betas,p):
    circuit = QuantumCircuit(n+3, n)
    for pk in range(p):
        for node in G.nodes():
            circuit.rz(-gammas[pk], node)
        for k, k_nbrs in G.adjacency():
            k_nbrs = list(k_nbrs.keys())
            k_nbrs = [int(x) for x in k_nbrs]
            circuit.x(k_nbrs)
            if len(k_nbrs)==1:
                circuit.crx(2betas[pk],k_nbrs[0],k)
            elif len(k_nbrs)==2:
                circuit.ccx(k_nbrs[0],k_nbrs[1],n)
                circuit.crx(2betas[pk],n,k)
                circuit.ccx(k_nbrs[0],k_nbrs[1],n)
            elif len(k_nbrs)==3:
                circuit.ccx(k_nbrs[0],k_nbrs[1],n)
                circuit.ccx(k_nbrs[2],n,n+1)
                circuit.crx(2betas[pk],n+1,k)
                circuit.ccx(k_nbrs[2],n,n+1)
                circuit.ccx(k_nbrs[0],k_nbrs[1],n)
            elif len(k_nbrs)==4:
                circuit.ccx(k_nbrs[0],k_nbrs[1],n)
                circuit.ccx(k_nbrs[2],n,n+1)
                circuit.ccx(k_nbrs[3],n+1,n+2)
                circuit.crx(2betas[pk],n+1,k)
                circuit.ccx(k_nbrs[3],n+1,n+2)
                circuit.ccx(k_nbrs[2],n,n+1)
                circuit.ccx(k_nbrs[0],k_nbrs[1],n)
            circuit.x(k_nbrs)
```

Assignees

No one assigned

Labels

No labels

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications

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Participants



```

circuit.measure(range(n),range(n))
return circuit

p = 1
gammas = np.array([3.96])
betas = np.array([4.70])

circuit = produce_QAOA_state(gammas, betas, p)
simulator = AerSimulator(method='matrix_product_state')
tcirc1 = transpile(circuit, simulator)
result1 = simulator.run(tcirc1).result()
counts1 = result1.get_counts(0)

simulator = AerSimulator(method='statevector')
tcirc2 = transpile(circuit, simulator)
result2 = simulator.run(tcirc2).result()
counts2 = result2.get_counts(0)

```

What is the expected behavior?

They should output similar measurement results.

Suggested solutions

I have no idea.

[Matrix Product State Simulation Method_test.zip](#)
[Matrix Product State Simulation Method_test.zip](#)

FEATURE REQUEST TEMPLATE

What is the expected behavior?

ENHANCEMENT REQUEST TEMPLATE

What is the expected enhancement?



merav-aharoni on May 19, 2021

Contributor ...

Hi @liyaochong , thanks for finding this bug!
The problem is with an optimization that is taking place - it is called `Fusion` . It should not be used together with MPS.
As a quick workaround, you can change the line:
`simulator = AerSimulator(method='matrix_product_state')` to
`simulator = AerSimulator(method='matrix_product_state', fusion_enable=False )` .
It should be disabled by default in the code. I will check why it is enabled in this case and will fix it.



liyaochong on May 19, 2021

Author ...

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Thank you for your reply! It's OK now!
By the way, Dose MPS simulator supports running on a GPU?



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By the way, Dose MPS simulator supports running on a GPU?



merav-aharoni on May 19, 2021

Contributor ...

Sorry, it is not supported on GPU yet.



liyaochong closed this as **completed** on May 20, 2021

liyaochong on May 20, 2021

Author ...

Sorry, it is not supported on GPU yet.

Thank you very much!



liyaochong on May 20, 2021

Author ...

It's OK now!



merav-aharoni mentioned this on May 23, 2021

[Disable Fusion when running on MPS #1255](#)

Disable Fusion when running on MPS #1255

<> Code

Merged chrisectlectic merged 6 commits into [Qiskit:main](#) from [merav-aharoni:issue1253](#) on Jun 15, 2021

Conversation 5 Commits 6 Checks 0 Files changed 6 +79 -9



merav-aharoni commented on May 23, 2021

Contributor

Summary

Fusion optimization does not work correctly on the MPS simulation method. Therefore should be disabled, even when set in the config.

Details and comments

See failure example in issue [#1253](#).



merav-aharoni added 4 commits 4 years ago

- Disabled fusion for MPS 4bc944e
- Added tests to make sure MPS does not use fusion 536a430
- Added release note 0b4a2b2
- Removed empty line ff0628c

merav-aharoni requested review from [chrisectlectic](#), [mtreinish](#) and [vvilpas](#) as code owners 4 years ago

merav-aharoni changed the title [WIP]-Disable-Fusion-when-running-on-MPS Disable Fusion when running on MPS on May 23, 2021



yaelbh commented on May 24, 2021

Contributor

May 24, 2021 at 7:24 AM GMT+2

If a user explicitly asked "MPS and enable fusion" then I'm not sure that you can just override the user's request. Maybe invoke an error, or at least print a warning that you override an explicit request.



merav-aharoni commented on Jun 3, 2021

Contributor Author

There are two problems with writing a warning:

- There is no mechanism for warnings in the C++ code.
 - The default set-up for the config is to set `fusion_enable = True`, so that even when the use does not set this field explicitly, they will get the warning.
- So overall, I prefer not to have a warning.



Reviewers

- [chrisectlectic](#) ✓
- [mtreinish](#) ●
- [vvilpas](#) ●

Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

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